

Predicting Quebec Winter Electricity Demand

1. Dataset Selection

For this project, we selected Hydro-Quebec's Electricity Consumption of Customers Taking Part in Local Demand Response Program dataset. The dataset contains information from three electrical substations in Montreal, with more than 64,000 hourly observations and 30 variables over the period from January 2022 to June 2024. The purpose of this project is to predict hourly electricity consumption in Quebec during winter.

The dataset contains several categories of variables that are integral to predicting electricity demand, including:

- Hourly electricity consumption
- Date and Time
- Weather conditions (Temperature, humidity, wind and precipitation)
- Number of connected customers

Our project will investigate how influential these factors are for predicting future electricity consumption.

As a supporting dataset, we plan to use Statistics Canada's Electric power generation: monthly receipts, deliveries, and availability. It contains monthly electricity information for Canadian provinces from January 2008 onwards. This dataset will primarily be used to provide a broader context on electricity patterns in Quebec and to determine whether these patterns are consistent at the provincial level and across Hydro-Quebec substations.

Dataset Source:

Hydro-Quebec Open Data - *Electricity consumption of customers taking part in a local demand response program*

<https://donnees.hydroquebec.com/explore/dataset/consommation-clients-evenements-pointe/information/>

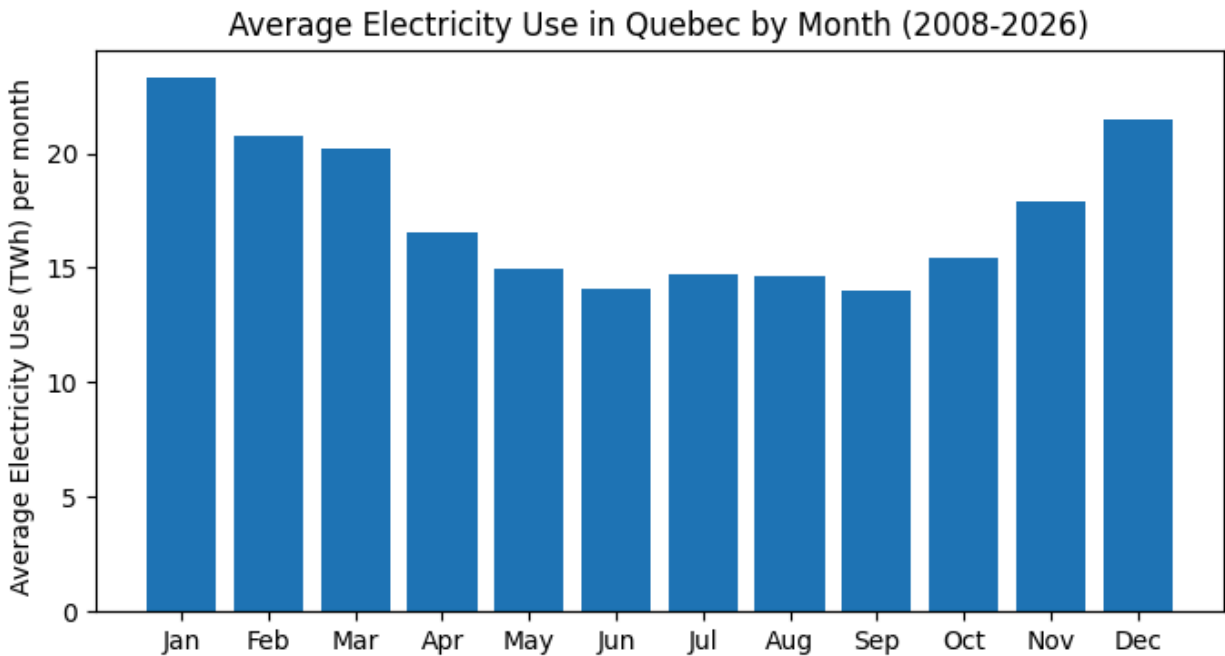
Statistics Canada's 25-10-0016-01 - *Electric power generation, monthly receipts, deliveries and availability*

<https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=2510001601>

2. Data Visualization and Initial EDA

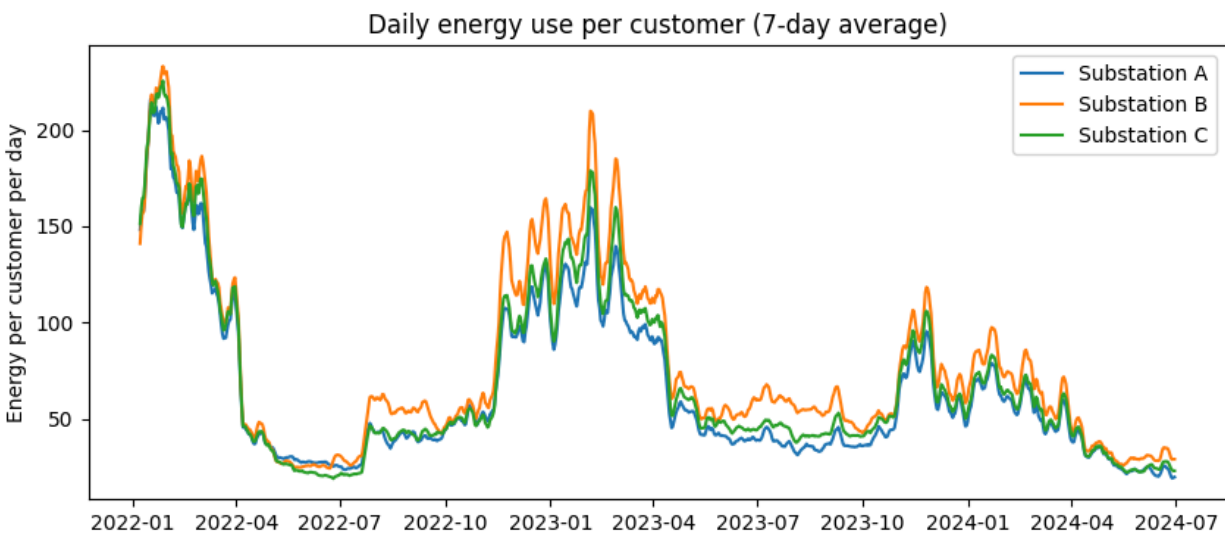
To get an idea of how overall energy consumption in Quebec has looked over the years, we set out some initial questions we wanted to answer.

1. What was the average energy consumption by month over the years?



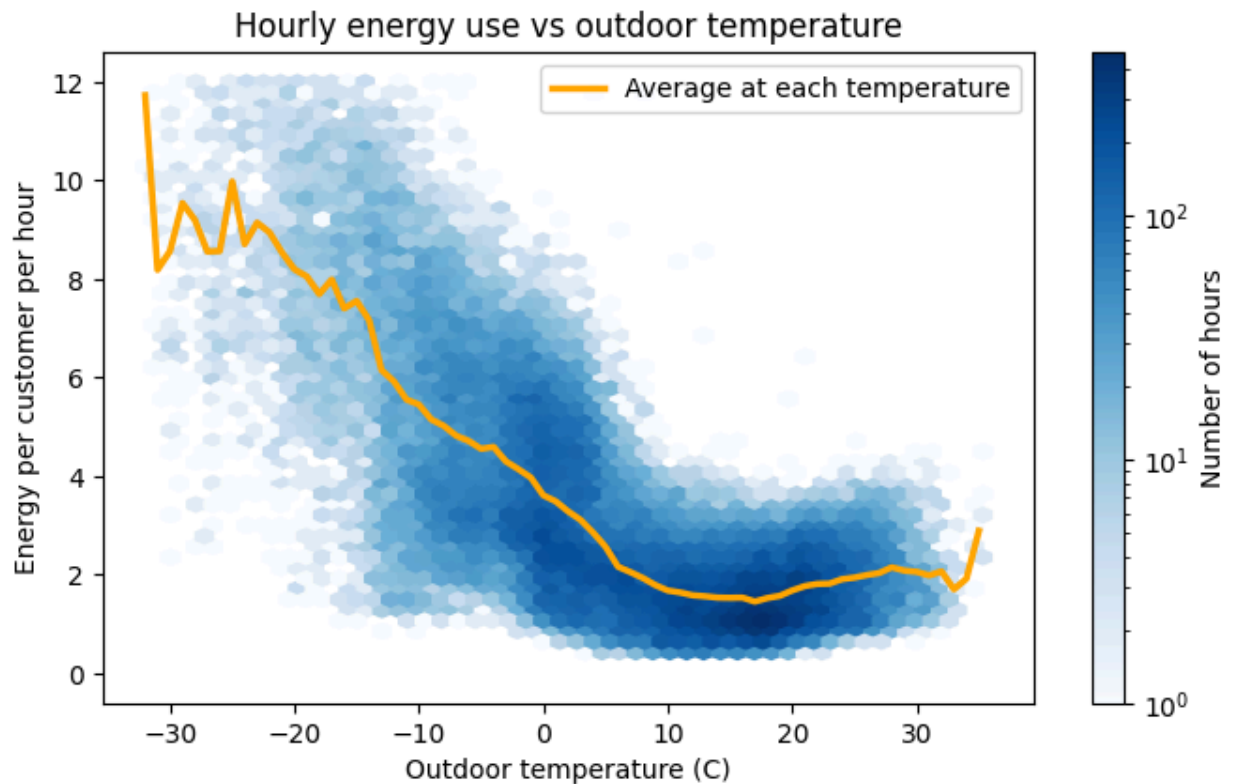
There is a trend over the years of higher energy consumption in the winter months, which makes sense given the demand for heating homes.

2. What was the energy use per customer by substation on a 7-day average?



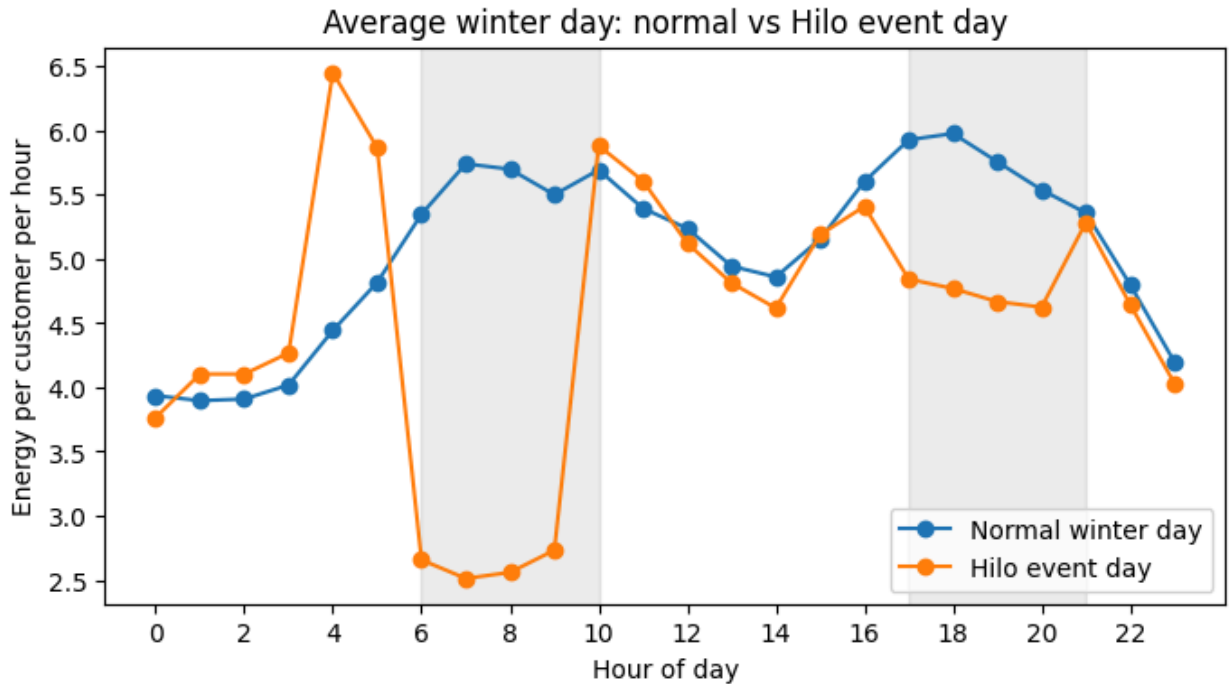
There seems to be a pattern of a peak in consumption during the winter months and a low peak in the summer months. Again showing a possible correlation with the demand for heating. There also seems to be a higher peak during the heavy winter months of January to March.

3. What was the average energy consumption per hour by the outdoor temperature?



There seems to be significant energy consumption as outdoor temperature decreases, with a slight increase as the temperature exceeds 20 Celsius.

4. What was the energy consumption by customer per hour during an average winter day vs a Hilo event day?



Hydro-Québec expects spikes and drops in energy consumption on Hilo event days, with most people choosing the morning challenge over the evening. There is a much bigger impact on the morning challenge participants than on the evening participants.

3. Dataset Filtering

Initial tidying was removing trailing spaces from the column names, dropping column 1 because it was just a row number added by the export and converting timestamps to Montreal time. We also filled the blank Hilo challenge types with “None” since they only appear when no event is running. We also found no duplicates.

Each substation was missing around 353 hours, and almost all of them were during midnight during daylight saving time. Each gap was a single hour, so we added the rows back, filled the values with the average of the neighbouring hours, rebuilt the calendar and holiday columns, and marked the rows with “was_missing”.

We also found some meter spikes that seemed to be due to logging issues. So we decided that any hour that was more than six times the average of its neighbours was replaced with that average and marked “spike_fixed,” thereby keeping the columns within realistic ranges. For StatCan, we just cleaned it up so it only had Quebec values.

4. Dataset Pre-processing

Our plan for pre-processing is to compare stations, since our target will be energy per customer. We will add a lag feature for consumption at 1 hour, 24 hours, and 1 week earlier, using only past values to avoid leaking future information. We will keep the existing sine/cosine time features and the weekend and holiday flags, turn the Hilo information into yes/no columns for pre-heating, event, and recovery, and one-hot encode the substation.

The data will be split by time, with training from January 2022 to June 2023, validation from July to December 2023, and testing from January to June 2024. Numeric features will be standardized using values from the set only for linear regression. The decision tree will use unscaled features, and we will also test whether dropping indoor temperature, which is highly correlated with outdoor temperature, affects accuracy.

5. Target and Task

Since this is a supervised regression task, given weather, calendar, customer, and demand-response information for an hour, the model predicts the energy consumption per customer at a substation for that hour. The reason we use per-customer rather than total is that, for example, substation C has many more customers than substation B, so using total energy would mostly reflect differences in customer size rather than actual energy demand. After cleaning, the target has a median of about 2.26 per customer per hour, and about 75% of hours are below 3.8.

6. Sample input and outputs

Example hours from substation A on Feb 7, 2023, a winter day with a morning hilo event, are available in the notebook.

From the notebook, it is clear that the temperature barely changes over the hours, but the output decreases significantly and then recovers.

7. Train, Validation and Test Strategy

We plan to split the data by time rather than randomly, so the models are always trained on earlier data and tested on later data. The neighbouring hours are very similar, so a random split would let the model see the test period during training, which can skew the results and make them look better than they should be. In practice, we plan to do this in scikit-learn's `train_test_split` with `shuffle=False`.

8. Primary Metric

Our primary metric will be mean absolute error, which measures how far each prediction is from the real value on average. We will also use root mean square error, which places greater weight on large errors, and mean absolute percentage error, as planned in our proposal. Results will be reported for all hours, winter hours, and Hilo event hours, and compared against a simple baseline that predicts each hour using the same hour from the day before.

9. Risks

A risk to consider is that this dataset covers a select group of customers who volunteered to participate in a local demand response program and were based in Montreal. Although this covers a core city environment in Quebec, it may not be the best model for more suburban or rural areas. Another risk with the participating demographic is how these people were selected. Hydro-Québec does not explicitly state the criteria by which these people were chosen, and there is a chance that the group does not best represent all energy customers in Montreal.

Another risk we could encounter would be the additional data from Statistics Canada on available energy, deliveries, and receipts. Since this data covers all of Quebec, it may differ slightly from our main dataset, which focuses on Montreal.

10. Feasibility and Plan

The cleaned dataset has about 65,000 rows and uses under 30 MB of memory, so it will run easily in other notebooks, including Google Colab. All work is done in a shared notebook that pulls data from our GitHub repository, so anyone can run and reproduce the results with ease.

The feasibility of our project appears promising; there is a positive correlation between weather conditions and energy consumption. We plan to dive deeper into this connection and, when modelling, see which conditions carry more weight in predicting energy consumption. We believe that once this balance is found, we should be able to make modest predictions. From here, we will evaluate if there is anything we can adjust or clean up in our data to improve the model.